

DECISION_TREE_IDS

April 20, 2026

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix
)

import warnings
warnings.filterwarnings("ignore")

plt.rcParams["figure.figsize"] = (10, 6)
sns.set_style("whitegrid")

[2]: file_path = "KDDTest+.arff"

attribute_names = []
data_started = False
data_rows = []

with open(file_path, "r", encoding="utf-8", errors="ignore") as f:
    for line in f:
        line = line.strip()

        # Skip empty lines and comments
        if not line or line.startswith("%"):
            continue

        # Read attribute names
```

```

if line.lower().startswith("@attribute"):
    parts = line.split()
    if len(parts) >= 2:
        attr_name = parts[1]
        attribute_names.append(attr_name)

# Detect beginning of data
elif line.lower() == "@data":
    data_started = True

# Read data rows
elif data_started:
    row = [x.strip() for x in line.split(",")]
    data_rows.append(row)

print("Total attributes:", len(attribute_names))
print("Total rows loaded:", len(data_rows))

```

Total attributes: 42
Total rows loaded: 22544

```

[3]: df = pd.DataFrame(data_rows, columns=attribute_names)

print("Dataset loaded successfully")
print("Shape:", df.shape)
df.head()

```

Dataset loaded successfully
Shape: (22544, 42)

```

[3]: 'duration' 'protocol_type' 'service' 'flag' 'src_bytes' 'dst_bytes' 'land' \
0      0      tcp      private      REJ      0      0      0
1      0      tcp      private      REJ      0      0      0
2      2      tcp      ftp_data      SF      12983      0      0
3      0      icmp      eco_i      SF      20      0      0
4      1      tcp      telnet      RST0      0      15      0

'wrong_fragment' 'urgent' 'hot' ... 'dst_host_srv_count' \
0      0      0      0 ...      10
1      0      0      0 ...      1
2      0      0      0 ...      86
3      0      0      0 ...      57
4      0      0      0 ...      86

'dst_host_same_srv_rate' 'dst_host_diff_srv_rate' \
0      0.04      0.06
1      0.00      0.06
2      0.61      0.04

```

3	1.00	0.00
4	0.31	0.17

	'dst_host_same_src_port_rate'	'dst_host_srv_diff_host_rate'	\
0	0.00	0.00	
1	0.00	0.00	
2	0.61	0.02	
3	1.00	0.28	
4	0.03	0.02	

	'dst_host_serror_rate'	'dst_host_srv_serror_rate'	'dst_host_rerror_rate'	\
0	0.00	0.00	1.00	
1	0.00	0.00	1.00	
2	0.00	0.00	0.00	
3	0.00	0.00	0.00	
4	0.00	0.00	0.83	

	'dst_host_srv_rerror_rate'	'class'
0	1.00	anomaly
1	1.00	anomaly
2	0.00	normal
3	0.00	anomaly
4	0.71	anomaly

[5 rows x 42 columns]

```
[4]: # Clean column names
df.columns = [col.strip().replace("'", "").replace('"', "") for col in df.
             ↪columns]

# Clean cell values
for col in df.columns:
    df[col] = df[col].astype(str).str.strip()
    df[col] = df[col].str.replace("'", "", regex=False)
    df[col] = df[col].str.replace('"', "", regex=False)

print("Columns and values cleaned successfully")
df.head()
```

Columns and values cleaned successfully

```
[4]: duration protocol_type service flag src_bytes dst_bytes land \
0      0          tcp private REJ      0          0      0
1      0          tcp private REJ      0          0      0
2      2          tcp ftp_data SF    12983          0      0
3      0          icmp eco_i SF      20          0      0
4      1          tcp telnet RST0      0         15      0
```

	wrong_fragment	urgent	hot	...	dst_host_srv_count	dst_host_same_srv_rate	\
0	0	0	0	...	10	0.04	
1	0	0	0	...	1	0.00	
2	0	0	0	...	86	0.61	
3	0	0	0	...	57	1.00	
4	0	0	0	...	86	0.31	

	dst_host_diff_srv_rate	dst_host_same_src_port_rate	\
0	0.06	0.00	
1	0.06	0.00	
2	0.04	0.61	
3	0.00	1.00	
4	0.17	0.03	

	dst_host_srv_diff_host_rate	dst_host_serror_rate	dst_host_srv_serror_rate	\
0	0.00	0.00	0.00	
1	0.00	0.00	0.00	
2	0.02	0.00	0.00	
3	0.28	0.00	0.00	
4	0.02	0.00	0.00	

	dst_host_rerror_rate	dst_host_srv_rerror_rate	class
0	1.00	1.00	anomaly
1	1.00	1.00	anomaly
2	0.00	0.00	normal
3	0.00	0.00	anomaly
4	0.83	0.71	anomaly

[5 rows x 42 columns]

```
[6]: print("Column Names:\n")
print(df.columns.tolist())

print("\nDataset Shape:")
print(df.shape)

print("\nMissing Values:\n")
print(df.isnull().sum())
```

Column Names:

```
['duration', 'protocol_type', 'service', 'flag', 'src_bytes', 'dst_bytes',
'land', 'wrong_fragment', 'urgent', 'hot', 'num_failed_logins', 'logged_in',
'num_compromised', 'root_shell', 'su_attempted', 'num_root',
'num_file_creations', 'num_shells', 'num_access_files', 'num_outbound_cmds',
'is_host_login', 'is_guest_login', 'count', 'srv_count', 'serror_rate',
'srv_serror_rate', 'rerror_rate', 'srv_rerror_rate', 'same_srv_rate',
'diff_srv_rate', 'srv_diff_host_rate', 'dst_host_count', 'dst_host_srv_count',
```

```
'dst_host_same_srv_rate', 'dst_host_diff_srv_rate',
'dst_host_same_src_port_rate', 'dst_host_srv_diff_host_rate',
'dst_host_serror_rate', 'dst_host_srv_serror_rate', 'dst_host_rerror_rate',
'dst_host_srv_rerror_rate', 'class']
```

Dataset Shape:
(22544, 42)

Missing Values:

duration	0
protocol_type	0
service	0
flag	0
src_bytes	0
dst_bytes	0
land	0
wrong_fragment	0
urgent	0
hot	0
num_failed_logins	0
logged_in	0
num_compromised	0
root_shell	0
su_attempted	0
num_root	0
num_file_creations	0
num_shells	0
num_access_files	0
num_outbound_cmds	0
is_host_login	0
is_guest_login	0
count	0
srv_count	0
serror_rate	0
srv_serror_rate	0
rerror_rate	0
srv_rerror_rate	0
same_srv_rate	0
diff_srv_rate	0
srv_diff_host_rate	0
dst_host_count	0
dst_host_srv_count	0
dst_host_same_srv_rate	0
dst_host_diff_srv_rate	0
dst_host_same_src_port_rate	0
dst_host_srv_diff_host_rate	0
dst_host_serror_rate	0

```
dst_host_srv_error_rate    0
dst_host_rerror_rate       0
dst_host_srv_rerror_rate   0
class                      0
dtype: int64
```

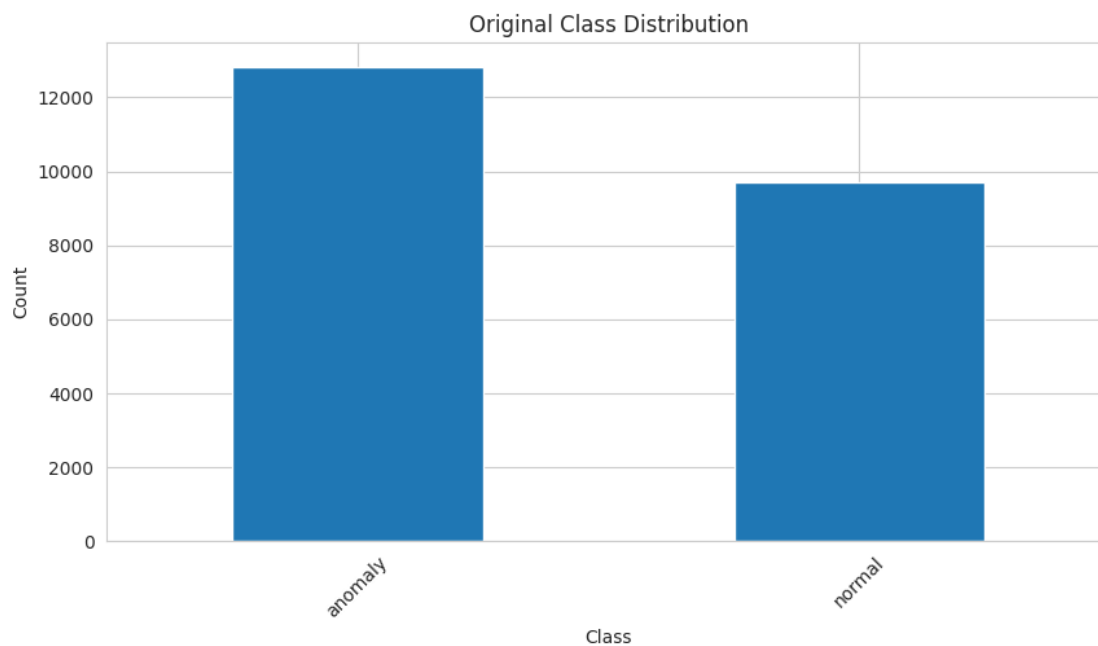
```
[7]: target_col = df.columns[-1]
      print("Target column:", target_col)

      print("\nUnique target classes:")
      print(df[target_col].unique())
```

Target column: class

Unique target classes:
<StringArray>
['anomaly', 'normal']
Length: 2, dtype: str

```
[8]: plt.figure(figsize=(10, 5))
      df[target_col].value_counts().plot(kind="bar")
      plt.title("Original Class Distribution")
      plt.xlabel("Class")
      plt.ylabel("Count")
      plt.xticks(rotation=45)
      plt.show()
```

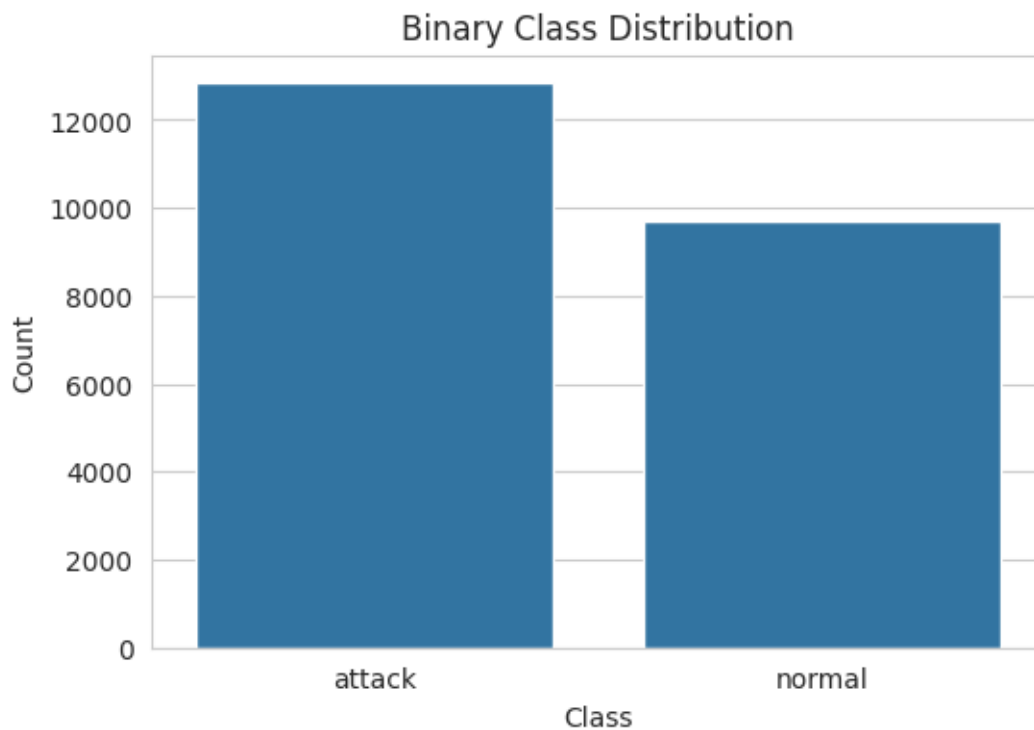


```
[9]: df["binary_class"] = df[target_col].apply(lambda x: "normal" if x == "normal"
      ↪else "attack")

print(df["binary_class"].value_counts())
```

```
binary_class
attack    12833
normal    9711
Name: count, dtype: int64
```

```
[10]: plt.figure(figsize=(6, 4))
sns.countplot(x="binary_class", data=df)
plt.title("Binary Class Distribution")
plt.xlabel("Class")
plt.ylabel("Count")
plt.show()
```



```
[11]: X = df.drop(columns=[target_col, "binary_class"])
y = df["binary_class"]

print("Feature shape:", X.shape)
print("Target shape:", y.shape)
```

```
Feature shape: (22544, 41)
```

Target shape: (22544,)

```
[12]: X = X.copy()

for col in X.columns:
    try:
        X[col] = pd.to_numeric(X[col])
    except:
        pass

print(X.dtypes)
```

duration	int64
protocol_type	str
service	str
flag	str
src_bytes	int64
dst_bytes	int64
land	int64
wrong_fragment	int64
urgent	int64
hot	int64
num_failed_logins	int64
logged_in	int64
num_compromised	int64
root_shell	int64
su_attempted	int64
num_root	int64
num_file_creations	int64
num_shells	int64
num_access_files	int64
num_outbound_cmds	int64
is_host_login	int64
is_guest_login	int64
count	int64
srv_count	int64
serror_rate	float64
srv_serror_rate	float64
rerror_rate	float64
srv_rerror_rate	float64
same_srv_rate	float64
diff_srv_rate	float64
srv_diff_host_rate	float64
dst_host_count	int64
dst_host_srv_count	int64
dst_host_same_srv_rate	float64
dst_host_diff_srv_rate	float64
dst_host_same_src_port_rate	float64


```
dst_host_srv_diff_host_rate    float64
dst_host_serror_rate          float64
dst_host_srv_serror_rate       float64
dst_host_rerror_rate           float64
dst_host_srv_rerror_rate       float64
dtype: object
```

```
[13]: categorical_cols = X.select_dtypes(include=["object"]).columns
print("Categorical Columns:", categorical_cols.tolist())
```

```
Categorical Columns: ['protocol_type', 'service', 'flag']
```

```
[14]: X_encoded = X.copy()
label_encoders = {}

for col in categorical_cols:
    le = LabelEncoder()
    X_encoded[col] = le.fit_transform(X_encoded[col].astype(str))
    label_encoders[col] = le

print("Categorical columns encoded successfully")
X_encoded.head()
```

```
Categorical columns encoded successfully
```

```
[14]:
```

	duration	protocol_type	service	flag	src_bytes	dst_bytes	land	\
0	0	1	45	1	0	0	0	
1	0	1	45	1	0	0	0	
2	2	1	19	9	12983	0	0	
3	0	0	13	9	20	0	0	
4	1	1	55	2	0	15	0	

	wrong_fragment	urgent	hot	...	dst_host_count	dst_host_srv_count	\
0	0	0	0	...	255	10	
1	0	0	0	...	255	1	
2	0	0	0	...	134	86	
3	0	0	0	...	3	57	
4	0	0	0	...	29	86	

	dst_host_same_srv_rate	dst_host_diff_srv_rate	\
0	0.04	0.06	
1	0.00	0.06	
2	0.61	0.04	
3	1.00	0.00	
4	0.31	0.17	

	dst_host_same_src_port_rate	dst_host_srv_diff_host_rate	\
0	0.00	0.00	

1	0.00	0.00
2	0.61	0.02
3	1.00	0.28
4	0.03	0.02

	dst_host_serror_rate	dst_host_srv_serror_rate	dst_host_rerror_rate \
0	0.0	0.0	1.00
1	0.0	0.0	1.00
2	0.0	0.0	0.00
3	0.0	0.0	0.00
4	0.0	0.0	0.83

	dst_host_srv_rerror_rate
0	1.00
1	1.00
2	0.00
3	0.00
4	0.71

[5 rows x 41 columns]

```
[15]: X_encoded = X_encoded.fillna(0)

print("Total missing values after handling:", X_encoded.isnull().sum().sum())
```

Total missing values after handling: 0

```
[16]: target_encoder = LabelEncoder()
y_encoded = target_encoder.fit_transform(y)

print("Target classes:", target_encoder.classes_)
```

Target classes: ['attack' 'normal']

```
[17]: X_train, X_test, y_train, y_test = train_test_split(
    X_encoded,
    y_encoded,
    test_size=0.3,
    random_state=42,
    stratify=y_encoded
)

print("Training set shape:", X_train.shape)
print("Testing set shape:", X_test.shape)
```

Training set shape: (15780, 41)

Testing set shape: (6764, 41)

```
[18]: dt_model = DecisionTreeClassifier(
        criterion="gini",
        max_depth=8,
        random_state=42
    )

    dt_model.fit(X_train, y_train)

    print("Decision Tree model trained successfully")
```

Decision Tree model trained successfully

```
[19]: y_pred = dt_model.predict(X_test)
```

```
[20]: accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    print("Accuracy :", round(accuracy, 4))
    print("Precision:", round(precision, 4))
    print("Recall   :", round(recall, 4))
    print("F1-Score :", round(f1, 4))
```

Accuracy : 0.979
Precision: 0.9873
Recall : 0.9636
F1-Score : 0.9753

```
[21]: print("Classification Report:\n")
    print(classification_report(y_test, y_pred, target_names=target_encoder.
        ↪classes_))
```

Classification Report:

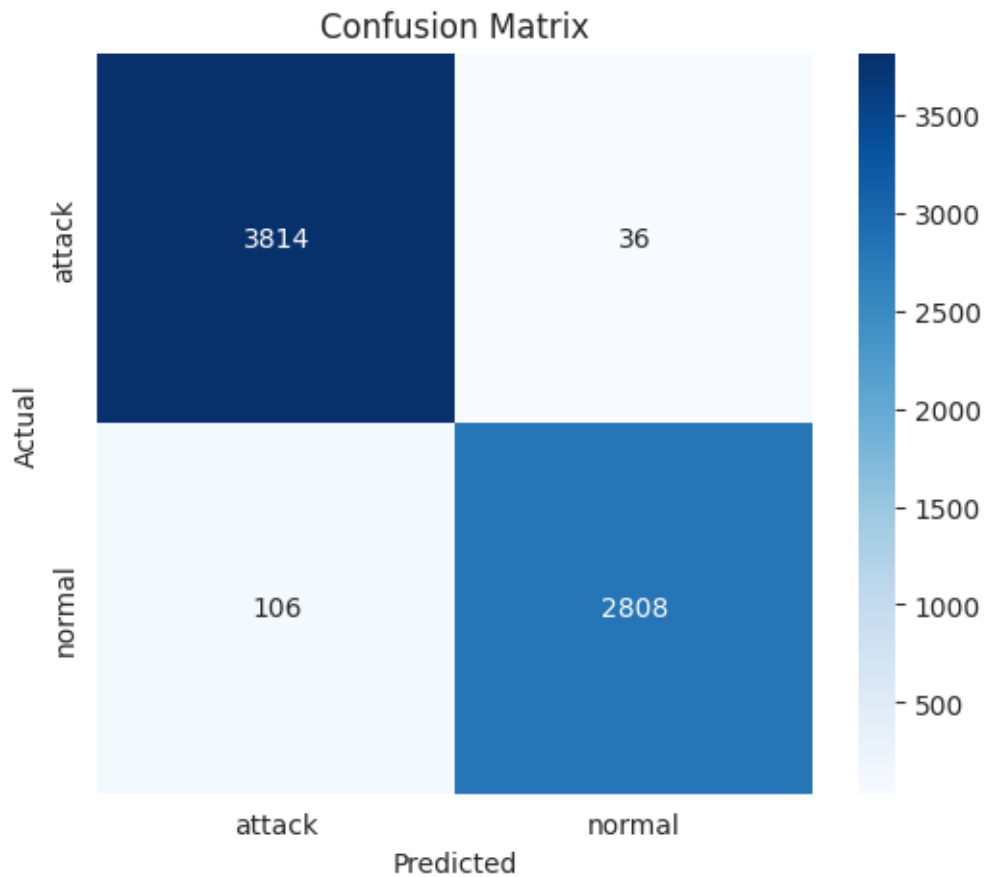
	precision	recall	f1-score	support
attack	0.97	0.99	0.98	3850
normal	0.99	0.96	0.98	2914
accuracy			0.98	6764
macro avg	0.98	0.98	0.98	6764
weighted avg	0.98	0.98	0.98	6764

```
[22]: cm = confusion_matrix(y_test, y_pred)
    print(cm)
```

```
[[3814  36]
```

```
[ 106 2808]]
```

```
[23]: plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=target_encoder.classes_,
    yticklabels=target_encoder.classes_
)
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



```
[24]: feature_importance = pd.DataFrame({
    "Feature": X_encoded.columns,
    "Importance": dt_model.feature_importances_
```

```

})

feature_importance = feature_importance.sort_values(by="Importance",
↪ascending=False)
feature_importance.head(10)

```

```

[24]:

```

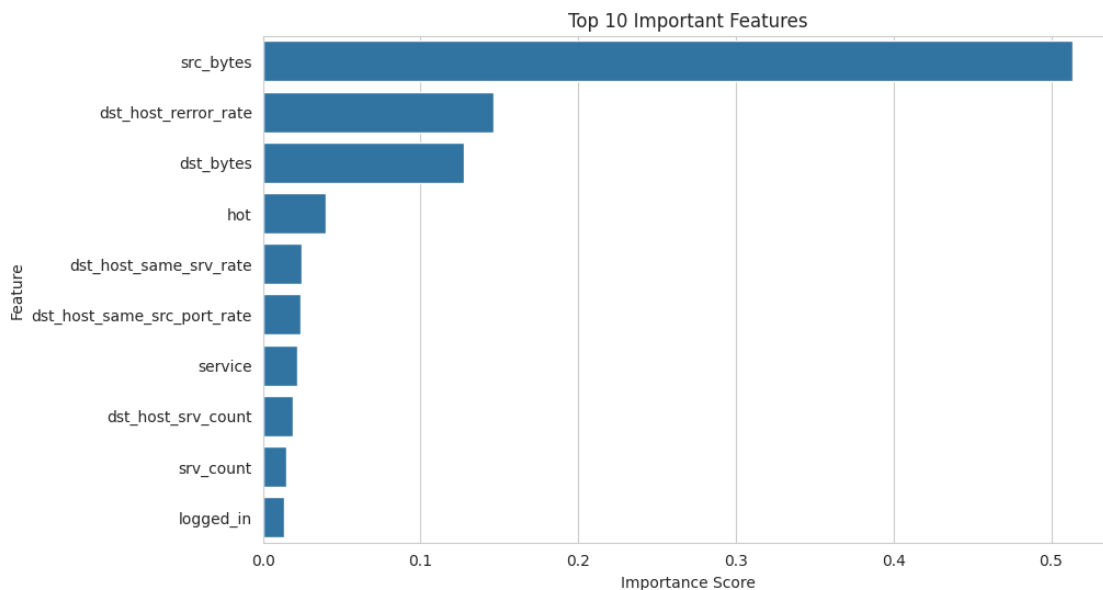
	Feature	Importance
4	src_bytes	0.513373
39	dst_host_rerror_rate	0.146552
5	dst_bytes	0.127237
9	hot	0.040061
33	dst_host_same_srv_rate	0.024512
35	dst_host_same_src_port_rate	0.023579
2	service	0.022004
32	dst_host_srv_count	0.018910
23	srv_count	0.014684
11	logged_in	0.013583

```

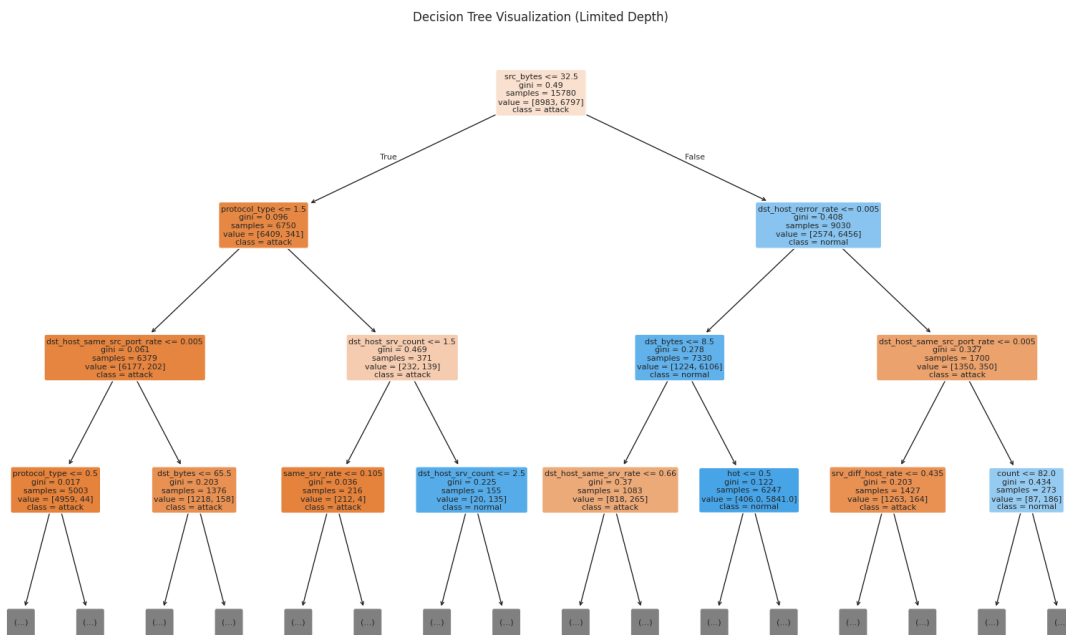
[25]: top_features = feature_importance.head(10)

plt.figure(figsize=(10, 6))
sns.barplot(x="Importance", y="Feature", data=top_features)
plt.title("Top 10 Important Features")
plt.xlabel("Importance Score")
plt.ylabel("Feature")
plt.show()

```



```
[26]: plt.figure(figsize=(20, 12))
plot_tree(
    dt_model,
    feature_names=X_encoded.columns,
    class_names=target_encoder.classes_,
    filled=True,
    rounded=True,
    fontsize=8,
    max_depth=3
)
plt.title("Decision Tree Visualization (Limited Depth)")
plt.show()
```



```
[27]: print("Result Summary")
print("-" * 40)
print(f"Accuracy : {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall : {recall:.4f}")
print(f"F1-Score : {f1:.4f}")
print("\nModel used: Decision Tree Classifier")
print("Task: Intrusion Detection (Binary Classification: normal vs attack)")
```

Result Summary

Accuracy : 0.9790

Precision: 0.9873
Recall : 0.9636
F1-Score : 0.9753

Model used: Decision Tree Classifier

Task: Intrusion Detection (Binary Classification: normal vs attack)